

The conjectured grid-gradient monotonicity can fail

Candidate counterexample for arXiv:2101.08108

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Status. Candidate counterexample, likely valid, pending expert review. This concerns only the fourth bullet of source Remark 4.2.

Source conjecture

Bottazzi studies the hyperfinite Neumann grid flow

$$u_t = \Delta_{\mathbb{X}}^* \phi(u) \tag{1}$$

and states four conjectural regularity properties in Remark 4.2. The final one asserts that the L^2 norm of $\mathbb{D}^{\pm*} \phi(u(t, \cdot))$ is nonincreasing in time [1, p. 18].

The entropy condition (6) is the grid function counterpart of an entropy condition that is classically used to single out physically relevant solutions to problem (4). Equation (7) states that $[u]$ and $[*\phi(u)]$ are a very weak solution to problem (4) (for the notion of very weak solution, see Lemma 5.3 of [13]).

Remark 4.2. *Numerical explorations of problem (5) suggest that its solution u satisfies further regularity conditions. In particular, we conjecture that*

- *if $\phi'(x) > 0$ for every $x \geq 0$, then for every initial data $\nu_{(t,x)}^{*\phi(u)}$ is Dirac for a.e. $(t, x) \in (0, +\infty) \times \Omega$;*
- *if $*\phi'(u(0, x)) > 0$ for a.e. $x \in \Omega_{\mathbb{X}}$, then $\nu_{(t,x)}^{*\phi(u)}$ is Dirac for a.e. $(t, x) \in (0, +\infty) \times \Omega$;*
- *for almost every initial data, there exists $\bar{t} \geq 0$ such that $\nu_{(t,x)}^{*\phi(u)}$ is Dirac for a.e. $(t, x) \in (\bar{t}, +\infty) \times \Omega$ (this is a trivial consequence of the previous point and of the asymptotic analysis carried out in Section 6 of [13]);*
- *the L^2 norm of $\mathbb{D}^{\pm*} \phi(u(t, \cdot))$ is nonincreasing in time.*

However, at this moment we only have been able to prove the first property. We sketch the proof of the first property and we discuss briefly the difficulties we encountered in the proof of the second. Denote by μ_L the Loeb measure induced by the product of the $$ Lebesgue measure over $*[0, +\infty)$ and the hy-*

Transcription of the target bullet. “The L^2 norm of $\mathbb{D}^{\pm*} \phi(u(t, \cdot))$ is nonincreasing in time.”

Counterexample

Define the standard function

$$\phi(s) = \begin{cases} 4s, & s \leq 0, \\ s(2-s)^2, & 0 \leq s \leq 2, \\ 0, & s \geq 2. \end{cases} \tag{2}$$

The values and first derivatives match at 0 and 2, so $\phi \in C^1(\mathbb{R})$ and is globally Lipschitz. It satisfies $\phi(0) = 0$ and $\phi(s) \geq 0$ for $s \geq 0$, as required in the source. On the middle interval,

$$\phi'(s) = (2-s)(2-3s), \quad (3)$$

which is strictly negative for $2/3 < s < 2$ and vanishes at $s = 2$.

Take $\Omega = (0, 1)$ and its source hyperfinite grid. Choose a nonzero $C_c^\infty((1/4, 3/4))$ bump, restrict its nonstandard extension to the grid, and call the resulting internal grid function ρ . After rescaling,

$$0 \leq \rho \leq 1, \quad \text{supp } \rho \subset [1/4, 3/4],$$

and set

$$u_0 = 2 - \frac{1}{2}\rho, \quad v_0 = {}^*\phi(u_0). \quad (4)$$

This is a nonnegative bounded initial datum satisfying source hypothesis (5'). It is already a grid function, so the initial projection fixes it. Its discrete derivatives have finite norms, and v_0 vanishes in a boundary layer.

Theorem 1. *For the source solution of (1) with (2) and (4), both functions*

$$t \mapsto \|\mathbb{D}^+ {}^*\phi(u(t, \cdot))\|_2, \quad t \mapsto \|\mathbb{D}^- {}^*\phi(u(t, \cdot))\|_2$$

have positive right derivative at $t = 0$. In particular, the fourth conjectural bullet of Remark 4.2 in [1] is false.

Proof Intuition

The expected Lyapunov calculation contains an extra factor that determines the sign. If $v = \phi(u)$, grid integration by parts gives

$$\frac{d}{dt} \frac{1}{2} \|\mathbb{D}v\|_2^2 = -\langle \phi'(u) \Delta_{\mathbb{X}} v, \Delta_{\mathbb{X}} v \rangle.$$

This is nonpositive on an increasing branch of ϕ , but it is *positive* on a decreasing branch. We place all nonconstant activity inside such a branch and make v zero near the boundary, eliminating every possible boundary-convention issue.

Proof

Put $v(t) = {}^*\phi(u(t))$ and

$$E_+(t) = \frac{1}{2} \|\mathbb{D}^+ v(t)\|_2^2.$$

Both v_0 and $v_t(0) = {}^*\phi'(u_0)u_t(0)$ vanish in a grid boundary layer: outside $\text{supp } \rho$ we have $u_0 = 2$, hence $v_0 = 0$ and $\phi'(u_0) = 0$. Thus the source Neumann Laplacian agrees on the relevant support with $\mathbb{D}^- \mathbb{D}^+$, and hyperfinite summation by parts has no boundary term. Differentiating exactly and using (1),

$$\begin{aligned} E'_+(0) &= \langle \mathbb{D}^+ v_0, \mathbb{D}^+ v_t(0) \rangle = -\langle \Delta_{\mathbb{X}} v_0, v_t(0) \rangle \\ &= -\langle \Delta_{\mathbb{X}} v_0, {}^*\phi'(u_0) \Delta_{\mathbb{X}} v_0 \rangle = -\int_{\Omega_{\mathbb{X}}} {}^*\phi'(u_0(x)) |\Delta_{\mathbb{X}} v_0(x)|^2 d\mathbb{X}. \end{aligned} \quad (5)$$

Where $\rho(x) > 0$, equation (4) gives $3/2 \leq u_0(x) < 2$, so (3) gives $\phi'(u_0(x)) < 0$. Where $\rho(x) = 0$, one has $u_0(x) = 2$ and $\phi'(u_0(x)) = 0$. Therefore every summand in the final expression in (5) is nonnegative.

At least one summand is strictly positive. Indeed, v_0 is nonzero and compactly supported in the connected grid. If $\Delta_{\mathbb{X}} v_0$ vanished at every point where $v_0 > 0$, applying the discrete maximum principle at a positive maximum and propagating equality along its positive component would force that component to meet a zero point with the same positive value, a contradiction. Hence

$$E'_+(0) > 0.$$

The initial gradient is nonzero, so its norm itself has positive right derivative. Replacing $\mathbb{D}^+$ by $\mathbb{D}^-$ gives the same energy by shifting the hyperfinite sum; boundary terms again vanish. This proves the theorem. $\square$

Exact five-node check

The sign is already visible with unit mesh on a five-node Neumann path. Take

$$u = (2, 2, 3/2, 2, 2), \quad v = \phi(u) = (0, 0, 3/8, 0, 0).$$

Then

$$\Delta v = (0, 3/8, -3/4, 3/8, 0), \quad \frac{1}{2} \|D^+ v\|_2^2 = \frac{9}{64},$$

and direct differentiation gives

$$E'_+(0) = - \sum_j \phi'(u_j) (\Delta v_j)^2 = \frac{45}{64} > 0.$$

The supplied verifier performs this calculation with exact rational arithmetic. Mesh scaling only multiplies the derivative by a positive power.

Verification, scope, and novelty

The hypotheses on ϕ and u_0 , the chain rule, the Neumann localization, both difference directions, and the strict sign were audited separately. This counterexample affects only the fourth bullet of Remark 4.2; the three Dirac-Young-measure bullets and the paper's main theorems are untouched.

A bounded search on 27 August 2026 checked the run indexes, official arXiv record, published Springer version, exact phrases from the conjecture, and erratum/correction queries. The published version retains the statement, and no explicit correction or counterexample was found. Novelty is plausible but remains provisional pending a broader expert literature review.

References

- [1] E. Bottazzi, *Describing limits of integrable functions as grid functions of nonstandard analysis*, Partial Differential Equations and Applications **2** (2021), article 51; arXiv:2101.08108.